

BEARDED DRAGON MANAGEMENT AND DISEASE: THE LATEST UPDATES

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Bearded dragons are the most common reptile pet in the United States and are popular because of their docile nature and small size. It is extremely important for the reptile veterinarian to be familiar with the husbandry requirements for this species along with the common and not so common medical conditions affecting them.

Several species of bearded dragons are native to Australia; the species common in the pet trade is the inland, or central, bearded dragon (*Pogona vitticeps*). They breed readily in captivity, and many color mutations and scale morphologies are available. Life expectancy is approximately 8-12 years. Bearded dragons are a member of the family Agamidae, and as such possess acrodont dentition and are not capable of tail autonomy. They are named for their large "beard" which can be inflated and turn very dark during offensive and defensive behaviors.

HUSBANDRY

Bearded dragons are native to arid woodland or scrubland habitat. They are a terrestrial, diurnal, sun-loving species. A 40- to 55-gallon aquarium or other similarly sized enclosure is the smallest appropriate sized habitat. They should be provided a thermal gradient with a hot basking area of 95-105°F (35-40°C) on one end and temperature of no lower than 80°F (26°C) on the cool end. Nighttime temperatures can drop to 73-75°F (22-24°C). Full spectrum lighting containing UVB (290-320 nm) should be provided 12 hours per day. A hide box should be provided, along with stacked flat stones or artificial plants for climbing and basking. Various substrates can be appropriate based on life stage and keeper experience. Bearded dragons are omnivores and prefer more insect matter as juveniles, but become more herbivorous as adults. A variety of gut-loaded insects such as crickets, mealworms, Dubia roaches, phoenix worms, etc. should be provided daily to juveniles and 1-2 times weekly for adults. Salads consisting of dark leafy greens and shredded vegetables should also be provided daily. Calcium and multivitamin supplement usage varies by life stage. A shallow water dish should be provided, though it is rare to see a bearded dragon drink. Soaking in shallow warm water should be done at least weekly, and many enjoy misting from a spray bottle and will learn to drink. Bearded dragons should typically be housed individually.

Bearded dragons in captivity in the United States will typically brumate in the fall starting in September or October. This period of reduced activity and appetite can last into the spring. Breeding season follows brumation, and animals in captivity will typically start breeding and laying eggs from January through June.

Females will readily cycle and lay eggs even without a male present.

ROUTINE MEDICAL CARE AND DIAGNOSTIC TESTING

A routine physical examination is recommended every 6-12 months. Observe the animal's posture and locomotion. They possess a parietal (third) eye on top of the head. Mucous membranes are normally pink to yellow, and the tip of the tongue is darker than the rest. The fat bodies, feces in the colon, and any developing follicles in females are readily palpable, but any other masses or firm areas should be investigated. Heart rate can be evaluated with a Doppler, and auscultation of the lungs can be performed by placing damp gauze between the head of the stethoscope and the animal's skin. Males have large femoral pores and hemipenial bulges in the base of the tail.

A fecal exam including a floatation and saline direct mount is recommended at least yearly as internal parasites are quite common in bearded dragons. The most commonly seen are pinworms (*Oxyuris* spp), coccidia (*Isospora amphiboluri*), and flagellated protozoans, most commonly *Trichomonas*. Some level of these organisms are considered normal in captive animals, but can cause disease if overly abundant. Cestodes and amoeba can rarely be seen as well. Acid-fast staining can be performed for *Cryptosporidium* if suspicion exists. A cloacal wash can be performed if a fecal sample is not readily available.

Venipuncture is easily accomplished in bearded dragons from the ventral coccygeal vein. A handler restrains the animal in a vertical position, pinning the legs if necessary. A 25-gauge needle on a tuberculin syringe is inserted on ventral midline and advanced until hitting bone, then slowly withdrawn while aspirating. 0.5-0.8% of body weight is a safe volume that can be taken from a healthy animal.

Radiographs and ultrasound are commonly used diagnostic tests. Radiographs can usually be easily obtained with minimal restraint. The patient will typically sit on the table without sedation for a dorsoventral view. The tube head can be rotated to take a lateral view with the animal sitting in the same position, or the patient can be rotated onto its side and tape or manual restraint used to pull the legs cranially and caudally. For ultrasound examination the animal can either be held vertically or in dorsal recumbency. Alcohol is used on the skin first to wet the area between scales, then ultrasound gel applied.

Endoscopy can also be used if additional diagnostic testing is needed. Bearded dragons are small enough that a rigid endoscope can be used for virtually all procedures. Cloacoscopy, tracheoscopy, and gastroscopy can all be easily performed. Celioscopy can be performed as well by making an incision along the flank to insert the endoscope and insufflating the coelom with room air or CO₂. Besides visual examination, samples can be collected for biopsy and culture, and topical treatments can be applied.

MEDICAL CONDITIONS

Atadenovirus (Adenovirus)

Atadenovirus is endemic in the US bearded dragon population. It is usually carried by adults, who are asymptomatic, and spread to hatchlings in which serious disease and high mortalities can result. Transmission is primarily fecal-oral, but vertical transmission is strongly suspected. Symptoms can be nonspecific such as lethargy, anorexia, stunted growth, and failure to thrive, but it can also cause diarrhea and opisthotonus. Secondary bacterial and parasitic infections are common. The virus typically causes an enteritis and hepatitis, but can cause encephalitis as well.¹ Diagnosis is by PCR on cloacal swabs or fecal samples, or on biopsy or necropsy. There is no specific treatment for the virus. Treatment is aimed at providing supportive care and treating any secondary infections. Prognosis is usually guarded to poor.

Internal Parasites

Internal parasites are extremely common in bearded dragons. Oxyurids and flagellated protozoa are normal inhabitants of the bearded dragon GI tract, and coccidia (*I. amphiboluri*) is endemic in the US population. All three are considered normal in small amounts, and if the bearded dragon is clinically normal then often no treatment is needed. However, parasite overgrowth is very common with concurrent stress and can cause illness. Oxyurids are the least likely to cause disease, but in large numbers can cause anorexia, unthriftiness, impaction, and cloacitis.¹ Treatment is with fenbendazole. The most common symptoms of protozoal enteritis are diarrhea, foul odor of the feces, blood in the feces, and sometimes anorexia or regurgitation. Treatment is with Metronidazole. Coccidiosis is especially common in young bearded dragons and often causes lethargy, anorexia, failure to thrive, diarrhea, and blood in the stool. Ponazuril at 30mg/kg PO q48h for 3 doses is the current preferred treatment.² A novel visceral coccidian has also been reported to affect the kidneys.³

Microsporidia

Microsporidians are obligate intracellular parasites, and can affect bearded dragons of all ages.^{1,4} It seems to be very common in large breeding facilities, and is transmitted by the fecal-oral route and possibly vertically as well. Signs include anorexia, lethargy, unthriftiness, swelling in affected areas, and acute death. Diagnosis is often made at necropsy. It can affect any body system, but has been reported in the eyelids, joints, liver, kidneys, intestine, and many others.^{1,4-6} Oral fenbendazole and topical fumagillin have been suggested as a treatment, but in most cases prognosis is poor.⁵

Yellow Fungus (CANV)

Fungal dermatitis is common in bearded dragons, and is often caused by the organism *Chrysosporium anamorph of Nannizziopsis vriesii* (CANV). This fungus typically causes raised, thickened, crusty lesions on the skin, and is also referred to as "yellow fungus." The fungus can also spread systemically.

Diagnosis is by biopsy, culture, and/or PCR. Treatment involves oral and topical antifungal medication with Voriconazole as the current treatment of choice. Prognosis is guarded to poor. Often lesions will improve, but will return when treatment is discontinued.

Nutritional Secondary Hyperparathyroidism (NSHP)

NSHP results from calcium deficiency caused by either a lack of calcium in the diet or lack of UVB lighting resulting in hypovitaminosis D. Calcium is leached from the bones resulting in soft, bowed long bones, fractures, deformities, stunted growth, weakness, anorexia, and eventually tremors, seizures, and death. History and exam findings are usually sufficient for diagnosis, but bloodwork or radiographs can be helpful as well. Treatment involves oral and parenteral calcium, parenteral vitamin D and calcitonin in some cases, along with correction of husbandry and supportive care. Prognosis depends on the severity of signs at presentation. Recovery can take weeks to months, and many patients will be left with permanent deformities.

Obesity and Hepatic Lipidosis

Captive bearded dragons are very commonly overfed, or fed insects high in fat; as such, obesity is very common. Fat is stored in two fat bodies in the coelomic cavity that are anchored caudally. Obesity can lead to other medical problems such as hepatic lipidosis, dystocia, difficulty swallowing and breathing, and difficulty moving. Dragons with hepatic lipidosis often present with nonspecific signs such as anorexia and lethargy, but are obviously overweight. Biochemistries can be normal, but can also demonstrate anemia, leukocytosis, and elevated AST or bile acids. Adult dragons should be offered vegetable matter daily and insects only 1-2 times per week. Avoid high-fat insects, and also encourage time out of the cage for exercise. Dragons with hepatic lipidosis often need syringe feeding and supportive care for weeks to months, along with hepatic support medications.

Periodontal Disease

Bearded dragons possess acrodont dentition in which the teeth are fused to the bone. The attachment site is covered by a thin layer of epithelial rather than gingival tissue, which predisposes to periodontal disease, particularly when fed soft foods. Clinical signs range from staining to erythema and calculus accumulation, to osteomyelitis, abscesses, bone loss, and sepsis. Treatment involves radiographs, scaling of the calculus, debridement of diseased bone, and antibiotic use based on culture and sensitivity. The teeth can be brushed for prevention using a cotton-tipped applicator with dilute chlorhexidine solution.

Impaction and Constipation

Constipation is seen most often in adult dragons due to chronic dehydration, lack of exercise, and improper diet, but can also be seen in any age secondary to substrate ingestion, NSHP, internal parasitism, intestinal lesions (eg, masses), or renomegaly. Dragons who are simply constipated have usually been

eating normally without producing feces, then the appetite declines and lethargy results. Typically this can be resolved with warm water soaks, belly massage, and an enema, and they remarkably improve within 24-48 hours of defecating. Animals with impactions due to other causes often display more severe symptoms. Similar treatment is attempted, and often syringe feeding small meals of a liquid high fiber formula can help as well. The underlying cause must also be addressed.

Renal Disease

Renal disease is common in aged bearded dragons, but other factors such as chronic dehydration or excessive dietary protein can also contribute. Common symptoms include lethargy, anorexia, weight loss, and constipation if renomegaly is present. A CBC and chemistry panel will often show elevated uric acid, phosphorous, potassium, or sodium levels with anemia and sometimes hypocalcemia. Articular and/or visceral gout can also develop with hyperuricemia. Aspirates of swollen joints will often demonstrate uric acid crystals.

Treatment involves supportive care with parenteral fluids, syringe feeding, and medications such as allopurinol and aluminum hydroxide. Long-term prognosis is poor.

Egg Retention

Reptiles can develop either pre- or post-ovulatory egg stasis due to hypocalcemia, improper husbandry, and failure to provide an appropriate nest site. Radiographs and ultrasound are useful to aid in diagnosis. Pre-ovulatory stasis should be treated with ovariectomy, otherwise progression to egg yolk coelomitis is common. Post-ovulatory stasis can be treated with oxytocin and provision of a nest box, though sometimes ovariectomies are necessary. Often supportive care with parenteral fluids, syringe feeding, and antibiotics are necessary before and after surgery.

Firefly Toxicosis

Fireflies contain lucibufagins, which are toxic and cause severe gastrointestinal upset, cardiac toxicity, and acute death. One firefly can be enough to kill an adult dragon. These animals typically present completely obtunded, nonresponsive, with extreme bradycardia. The author has heard of a case of recovery with intensive supportive care for days, but typically affected animals do not survive.

Aneurysm

Aneurysms are a more recently described syndrome and usually arise from the internal carotid artery or aorta.⁷ These usually present as a swelling on the head or neck with no other symptoms, but affected dragons can also be pale, lethargic, and weak. Surgical removal is challenging; ultrasound or MRI can be used to determine the origin of the aneurysm so that it can be ligated and removed, but a blood donor should be available.

Gastric Endocrine Carcinoma

Gastric endocrine carcinoma has been recognized more and more in the last 6-7 years. It tends to occur in young bearded dragons and can present with signs of anorexia, lethargy, regurgitation, and marked hyperglycemia on chemistries. Radiographs, ultrasound, and gastroscopy can be used to aid diagnosis. Celiotomy with gastrotomy is the treatment of choice, but prognosis is guarded to poor as these tumors are typically large or have metastasized to the liver by the time of diagnosis.^{1,8} These tumors produce somatostatin causing the hyperglycemia in many cases.⁸

Squamous Cell Carcinoma

While squamous cell carcinoma can be present anywhere, in bearded dragons it appears to be most prevalent in the eyelids or periocular tissues.⁹ It can appear as a solid mass or as an ulcerated wound. Diagnosis is by biopsy, and treatment involves surgical excision with wide margins, possibly enucleation. Metastasis is rare, and surgical excision appears to be curative if the entire tumor is removed.⁹

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